From boost data to candidate translations
A mass-family construction for the DSSYK singlet

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Abstract

We construct positive-energy Poincare representations extending the leading singlet boost kernel. Every positive mass is allowed, while an explicit translated overlap distinguishes the choices. A finite-chord seed calculation separates the microscopic Liouvillian from the length-preserving generators used in the semiclassical chord algebra. Two-particle composition then identifies the extra dynamical input needed to select a physical mass spectrum. The construction uses standard conformal mechanics and light-front kinematics; it supplies a testable family, not a completed microscopic translation dictionary.

1 A concrete extension of the boost kernel

Our microscopic input is the conserved-subtracted derivative singlet X of the preceding notes, normalized by $\tau(X^2) = 1$. The state is the full annealed normalized trace at $\beta = \mu_{\rm chem} = 0$. First take the complex DSSYK double scaling limit at $\lambda = p^2/N > 0$, then $\lambda \to 0$ at fixed string time. Its kernel becomes

$$C(\eta) = \operatorname{sech}^4(\eta/2), \qquad \eta = 4\mathcal{J}_s t. \tag{1}$$

We use the candidate boost clock η and $\hbar = c_{\rm light} = 1$. On $\mathcal{H} = L^2(\mathbb{R}_+, \mathrm{d}p)$, with dimensionless p, define

$$Q = p, \qquad B = i(p\partial_p + \tfrac{1}{2}), \qquad \psi_d(p) = \frac{2^d}{\sqrt{\Gamma(2d)}} p^{d-1/2} e^{-p}, \qquad d = 2. \tag{2}$$

The dilation group is unitary and acts as

$$(e^{-i\eta B} f)(p) = e^{\eta/2} f(e^{\eta} p), \qquad \langle \psi_d | e^{-i\eta B} | \psi_d \rangle = \operatorname{sech}^{2d}(\eta/2). \tag{3}$$

Thus it realizes Eq. (1) exactly. For arbitrary $m, \mu > 0$, both with energy units, set

$$\boxed{P^+ = \mu Q, \qquad P^- = \frac{m^2}{\mu Q}, \qquad P^0 = \frac{P^+ + P^-}{2}, \quad P^1 = \frac{P^+ - P^-}{2}.} \tag{4}$$

Here $P^\pm = P^0 \pm P^1$, with no $\sqrt{2}$. On the common invariant core $C_c^\infty(0, \infty)$,

$$[B, P^0] = iP^1, \quad [B, P^1] = iP^0, \quad [P^0, P^1] = 0, \quad (P^0)^2 - (P^1)^2 = m^2, \quad P^0 \geq m. \tag{5}$$

The translations are self-adjoint multiplication operators on their maximal domains. These are the standard positive-energy light-front relations [3], now attached to the explicit singlet-shaped state. The boost data alone have not selected $m/\mathcal{J}_s$ or $\mu/\mathcal{J}_s$.

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2 Why the conformal label does not fix this mass

Complete Q, B to a standard positive discrete-series realization:

$$S = -p\partial_p^2 - \partial_p + \frac{(d-1/2)^2}{p}, \qquad E = \frac{Q+S}{2}, \quad \Pi = \frac{Q-S}{2}. \tag{6}$$

The self-adjoint discrete-series realization at $d = 2$ obeys

$$\begin{aligned} [B, E] &= i\Pi, & [B, \Pi] &= iE, & [E, \Pi] &= iB, \\ E^2 - \Pi^2 - B^2 &= d(d-1), & E\psi_d &= d\psi_d. \end{aligned} \tag{7}$$

This is a change of variable and measure in conformal mechanics [2]. For example, its positive multiplication generator $K = x^2/2$ becomes $Q = p$ under $p = x^2/2$; the kinetic generator becomes S with representation parameter $m_{\mathrm{A}} = 2d - 1$. This source parameter is unrelated to our Poincare mass m.

In Eq. (4), the second null translation is $m^2/(\mu Q)$, *not* μS. The latter contains derivatives and does not commute with Q: $[Q, S] = -2iB$. Consequently the conformal Casimir and the Poincare mass are different invariants. Every $m > 0$ in Eq. (4) uses the same $d = 2$ conformal representation and state.

There is also a direct Hilbert-space interpretation. Under $p = e^y$, the unitary map $f \mapsto g(y) = e^{y/2} f(e^y)$ gives

$$B = i\partial_y, \qquad g_d(y) = \frac{2^d}{\sqrt{\Gamma(2d)}} e^{dy - e^y}. \tag{8}$$

Its Fourier transform is proportional to $\Gamma(d - ik)$, which has no zeros for real k. The boost orbit of g_d is cyclic. Equality of the positive kernels therefore gives an isometry between the limiting cyclic space generated by $X(t)$ and this boost representation. It does not specify the action of translations on products of microscopic operators, nor an affiliation with a physical wedge algebra.

To see the ordinary particle content, set $p = (m/\mu)e^\theta$ and $g(\theta) = \sqrt{p} f(p)$. Then

$$\mathcal{H} \simeq L^2(\mathbb{R}, \mathrm{d}\theta), \qquad B = i\partial_\theta, \quad P^0 = m\cosh\theta, \quad P^1 = m\sinh\theta. \tag{9}$$

Changing μ shifts the rapidity origin. Different m give inequivalent Poincare mass sectors despite identical boost-only overlaps.

For comparison, the usual *linear* contraction sets $P_R^0 = E/R$, $P_R^1 = \Pi/R$, keeping B fixed:

$$[P_R^0, P_R^1] = iB/R^2, \qquad (P_R^0)^2 - (P_R^1)^2 = \frac{d(d-1) + B^2}{R^2}. \tag{10}$$

On a fixed vector with finite generator norms, translations become trivial as $R \to \infty$. Massive representation contractions require a family of representations/states and identifications; such contractions are established mathematics [4]. The nonlinear completion in Eq. (4) is not that contraction, and it introduces m explicitly.

3 A translated kernel that can distinguish masses

Consider the ordered overlap

$$K_{m,\mu}(a,b;\eta;\epsilon) = \langle\psi_d|e^{-(\epsilon+ia)P^0+ibP^1}e^{-i\eta B}|\psi_d\rangle, \qquad \epsilon \geq 0. \tag{11}$$

Here a, b are inertial time and space displacements, with inverse-energy units; they are not microscopic string time t. Direct substitution gives

$$\begin{aligned} \alpha &= 1 + e^{\eta} + \tfrac{\mu}{2}[\epsilon + i(a-b)], \qquad & \gamma &= \tfrac{m^2}{2\mu}[\epsilon + i(a+b)], \\ K_{m,\mu} &= \frac{2^{2d}e^{d\eta}}{\Gamma(2d)}\int_0^\infty p^{2d-1}e^{-\alpha p-\gamma/p}\mathrm{d}p \\ &= \frac{2^{2d+1}e^{d\eta}}{\Gamma(2d)}\left(\frac{\gamma}{\alpha}\right)^d K_{2d}(2\sqrt{\alpha\gamma}). \end{aligned} \tag{12}$$

The last K_{2d} denotes the modified Bessel function. At $\gamma = 0$ use the integral's regular limit $2^{2d}e^{d\eta}\alpha^{-2d}$. For $\epsilon > 0$ the standard Laplace integral applies; dominated convergence gives the unitary $\epsilon \downarrow 0$ boundary value. The Bessel branches are continued from that region. At $a = b = \epsilon = 0$, all masses give Eq. (3).

For $d = 2$, $\mu = m$ and $a = 0.5$, $b = \eta = \epsilon = 0$, in a chosen common energy unit, one instead obtains

m	$\mathrm{Re}\,K$	$\mathrm{Im}\,K$
0.5	0.94098725	−0.32553034
1.0	0.77393294	−0.60636054
2.0	0.23436204	−0.90741504

These are candidate predictions, not measured DSSYK translation data. One independently identified translation observable would distinguish this family in a way that another measurement of the same boost kernel cannot.

The state has exact moments

$$\langle p^r\rangle = 2^{-r}\frac{\Gamma(2d+r)}{\Gamma(2d)}, \qquad r > -2d. \tag{13}$$

In particular, for $d = 2$, $\mu = m$, $\langle P^0\rangle = 4m/3$ and $\langle (P^0)^2\rangle = 23m^2/12$. The fourth energy moment diverges logarithmically at $p = 0$. Thus ψ_2 belongs to $D(P^0)$ but not to $D((P^0)^2)$. The mass identity on the core, and its quadratic-form version on this state, remain valid; one should not manipulate four energy derivatives of Eq. (11) as if all moments existed.

4 An exact seed test for the chord generators

Lin–Stanford define the length-preserving generators [1]

$$J_{ij} = a_i^\dagger a_j - [\bar{n}]_q, \qquad c = q^{\bar{n}/2},$$
$$E_{\rm LS} = -\frac{J_{LL}+J_{RR}}{2c}, \qquad B_{\rm LS} = \frac{J_{LL}-J_{RR}}{2c}, \qquad P_{\rm LS} = \frac{i}{2}(J_{LR}-J_{RL}). \tag{14}$$

We use $h = 4\lambda$, $q = e^{-h}$ and $d = 2$ for the singlet chord. At the tracial insertion seed $|d;0,0\rangle$, both lowering operators annihilate the state and $\bar{n} = d$. Hence, exactly at any $h > 0$,

$$J_{ij}|d;0,0\rangle = -[d]_q|d;0,0\rangle, \quad B_{\rm LS}|d;0,0\rangle = P_{\rm LS}|d;0,0\rangle = 0, \quad E_{\rm LS}|d;0,0\rangle = \frac{[d]_q}{q^{d/2}}|d;0,0\rangle. \tag{15}$$

Rescaling $\widehat{B}_{\rm LS} = B_{\rm LS}/\sqrt{1-c^2}$ and similarly $\widehat{P}_{\rm LS}$ does not change these zeros. Consequently

$$\frac{\|([\widehat{B}_{\rm LS}, \widehat{P}_{\rm LS}] - iE_{\rm LS})|d;0,0\rangle\|}{\|E_{\rm LS}|d;0,0\rangle\|} = 1. \tag{16}$$

The smooth large-chord semiclassical $\mathfrak{sl}_2$ approximation cannot be made uniform on this fixed seed simply by taking $h \to 0$. Its source derivation assumes a large chord number and specified smooth states; a new state preparation/limit must be supplied here.

The microscopic boost candidate is instead the Liouvillian

$$B_{\rm tr} = \frac{H_L - H_R}{4\mathcal{J}_s} = \frac{H_{{\rm src},L} - H_{{\rm src},R}}{2}. \tag{17}$$

Using the one-chord Gram matrix, with off-diagonal entry q^d, gives

$$\|B_{\rm tr}|d;0,0\rangle\|^2 = \frac{1-q^d}{2h} \longrightarrow \frac{d}{2} = 1, \qquad \|\widehat{B}_{\rm LS}|d;0,0\rangle\|^2 = 0. \tag{18}$$

This agrees with $-C''(0) = 1$ in the boost variable. The two operators cannot be identified on the same seed. This test does not invalidate the earlier scalar four-point expansion: it concerns a proposed new operator identification, not those independently checked correlators.

The exact chord Casimir is $\Omega = q^d + q^{1-d}$. Its normalized limit

$$\lim_{h\to 0} \frac{\Omega - (1+q)}{h^2} = d(d-1) = 2 \tag{19}$$

records the representation label. No dimensionful translation scale or Poincare mass has thereby been fixed.

5 Two-particle composition is a further constraint

The one-particle rule $P^- = m^2/P^+$ must not be applied to a tensor product by simply replacing P^+ with its sum while keeping the same mass. For two freely propagating equal-mass particles, use the primitive sum $B_{12} = B_1 + B_2$ and $P^\pm_{12} = P^\pm_1 + P^\pm_2$. Then

$$M^2_{12} = m^2\left(2 + \frac{p_1}{p_2} + \frac{p_2}{p_1}\right) = \frac{m^2}{x(1-x)} \geq 4m^2, \qquad x = \frac{p_1}{p_1 + p_2}. \tag{20}$$

This is the standard free light-front invariant mass [3]. For the normalized symmetric product $\psi_d \otimes \psi_d$, p_1, p_2 are independent gamma variables with shape $k = 2d$ and rate 2; x has the Beta(k, k) distribution. Thus $s = M^2_{12}/m^2$ has density

$$\rho_d(s) = \frac{2}{\mathrm{B}(2d, 2d)\, s^{2d+1}\sqrt{1 - 4/s}}, \quad s > 4, \qquad \langle s \rangle_{d=2} = \frac{14}{3}. \tag{21}$$

This is a two-particle continuum, not a discrete meson tower. The construction shows where dynamics must enter: a microscopic choice of P^- (including its multiparticle action) must determine the invariant mass operator. The one-particle relation alone leaves its mass input free.

6 Verification and the next calculation

Nine exact differential identities verify Eqs. (7) and (5), including the lowest-weight state. The seed Gram calculation and the Beta normalization/mean are exact. Eight translated kernels were independently integrated on the real p axis at 45 and 65 digits, including both signs of the inverse-p phase, null displacements and positive damping. For $p \in [\delta, R]$, the omitted absolute tails obey

$$|\Delta K| \leq A\left[\frac{\delta^{2d}}{2d} + \frac{\Gamma(2d, (\mathrm{Re}\,\alpha)R)}{(\mathrm{Re}\,\alpha)^{2d}}\right], \quad A = \frac{2^{2d} e^{d\eta}}{\Gamma(2d)}. \tag{22}$$

We used $\delta = 0.0005$, $R = 64$. The largest absolute discrepancy against Eq. (12) is 1.14×10^{-13}, within the explicit tail bound; the largest relative discrepancy is 1.30×10^{-13}. Precision refinement changes the interior integral by less than 3.2×10^{-46}. This last check is empirical, not interval-certified. The cancellation upper bound is below 1.21 at these points.

There is a useful constraint on this search. In a finite tracial algebra, let a positive affiliated operator A obey exact covariance $\alpha_\eta(A) = e^\eta A$ under trace-preserving automorphisms. Its spectral tail $F(x) = \tau(1_{(x,\infty)}(A))$ then satisfies $F(x) = F(e^\eta x)$. Sending $\eta \to \infty$ gives $F(x) = 0$ for every $x > 0$, hence $A = 0$ by faithfulness. A nonzero null generator therefore cannot be such a one-sided observable with exact covariance. Two-sided GNS operators or a singular change of the limiting algebra/state are possible routes; approximate finite-h covariance is not excluded.

The next calculation should construct a microscopic positive null generator in this enlarged setting, including its action on two-singlet states, or an independent translation Ward identity. Its mass operator can then be compared with the candidate QCD_2 spectrum [5]. The present $d = 2$ temporal data do not select that generator. The finite-h four-point kernel is already available as a subsequent interaction test once this map is fixed.

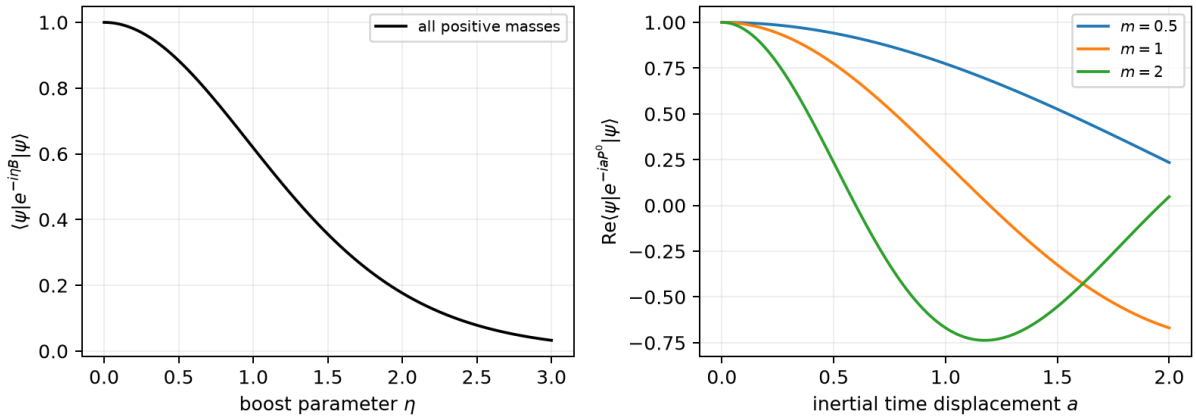


Figure 1: Identical boost data, distinct translated overlaps. The right panel uses $\mu = m$ and a common arbitrary energy unit. These curves are predictions of the constructed family, not a determination of the DSSYK mass spectrum.

The prior Rindler construction used a regulated spatial KG packet and the Minkowski vacuum restricted to a wedge, with a midpoint two-point function. Here the state is a normalized vector in a global positive-energy one-particle representation. Equality of the scalar boost overlap does not equate these preparation protocols. In particular, no ordinary thermal four-point contour has been imposed in the present construction.

All computations, source locations, dependency versions and hashes are saved with this local stage. No Git commit or publication accompanies it.

References

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